

JOSHUA FAGIN

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RESEARCH AREAS

Topics: Quasar variability, accretion disk structure, strong gravitational lensing, microlensing, dark matter substructure, time-domain astronomy

Methods: Bayesian modeling, machine learning, physics-informed neural networks, simulation-based inference, Markov chain Monte Carlo, Gaussian process regression

TECHNICAL SKILLS

Deep learning frameworks: PyTorch, Tensorflow/Keras, Tensorflow-Probability

Deep learning techniques: Transformers, latent SDEs/ODEs, RNNs, CNNs, variational autoencoders, normalizing flows, simulation-based inference

Data analysis: Python, NumPy, Matplotlib, SciPy, Scikit-Learn, Astropy, GPyTorch, BoTorch, emcee, Pyro, Pandas, Numba, AstroPy, Jupyter

Tools and environments: Linux, distributed training, HPC, Kubernetes, Slurm, GitHub, L^AT_EX

EDUCATION

CUNY Graduate Center

August 2020 - June 2025

Ph.D. in Physics (GPA: 3.9/4.0)

Advisor: Professor Matthew O'Dowd

Cornell University

August 2016 - May 2020

B.S. Physics, B.S. Mathematics

PROFESSIONAL EXPERIENCE

CUNY & American Museum of Natural History

Postdoctoral Researcher in Astrophysics and Machine Learning

July 2025 – Present

Ph.D. Researcher in Astrophysics and Machine Learning

August 2020 – June 2025

- Working on various projects with a focus on deep machine learning applied to time series of quasar variability, including microlensing of strongly lensed quasars, and images of strongly lensed galaxies.

Fermilab Muon g-2 Experiment, Cornell University

Undergraduate Researcher in Experimental Particle Physics

March 2018 – August 2020

- Worked with Professor David Rubin to develop a Fourier method for extracting the momentum distribution of the muon beam. Developed an analytic model and simulation of the muon signal.

PUBLICATIONS IN REFEREED JOURNALS

First Author

- **J. Fagin**, J. Chan, H. Best, M. O'Dowd, K. E. S. Ford, M. J. Graham, J. W. Park, V. A. Villar, *Joint Modeling of Quasar Variability and Accretion Disk Reprocessing using Latent Stochastic Differential Equations*, ApJ, 988, 59, 2025
- **J. Fagin**, E. Paic, F. Neira, H. Best, G. Vernardos, D. Sluse, T. Anguita, M. O'Dowd, M. Millon, *Predicting High-magnification Events in Microlensed Quasars in the Era of LSST using Recurrent Neural Networks*, ApJ, 981, 61, 2025
- **J. Fagin**, J. W. Park, H. Best, J. Chan, K. E. S. Ford, M. J. Graham, V. A. Villar, S. Ho, M. O'Dowd, *Latent Stochastic Differential Equations for Modeling Quasar Variability and Inferring Black Hole Properties*, ApJ, 965, 104, 2024
- **J. Fagin**, G. Vernardos, G. Tsagkatakis, P. Pantazis, A. J. Shajib, M. O'Dowd, *Measuring the Substructure Mass Power Spectrum of 23 SLACS Strong Galaxy-Galaxy Lenses with Convolutional Neural Networks*, MNRAS, 532, 2, 2024

Second Author

- J. Chan, **J. Fagin**, H. Best, M. O'Dowd, *Reverberation Mapping of Lamp-post and Wind Structures in Accretion Thin Disks*, ApJ, 978, 54, 2025
- H. Best, **J. Fagin**, G. Vernardos, M. O'Dowd, *Resolving the Vicinity of Supermassive Black Holes with Gravitational Microlensing*, MNRAS, 531, 1, 2024

Third Author

- H. Best, M. O'Dowd, **J. Fagin**, J. Chan, B. Ierace, *Amoeba: An AGN Model of Optical Emissions Beyond steady-state Accretion discs*, MNRAS, 539, 2, 2025

Ph.D. Thesis

- *Novel Machine Learning Methods for Quasar Variability and Strong Gravitational Lensing*

MACHINE LEARNING WORKSHOP PAPERS

- **J. Fagin**, J. Chan, H. Best, M. O'Dowd, K. E. S. Ford, M. J. Graham, J. W. Park, V. A. Villar, *Joint Modeling of Quasar Variability and Accretion Disk Reprocessing using Latent Stochastic Differential Equations*, ICML 2025 Machine Learning for Astrophysics Workshop
- **J. Fagin**, J. W. Park, H. Best, J. Chan, K. E. S. Ford, M. J. Graham, V. A. Villar, S. Ho, M. O'Dowd, *Latent Stochastic Differential Equations for Modeling Quasar Variability and Inferring Black Hole Properties*, ICLR 2023 Physics for Machine Learning Workshop
 - **Awarded spotlight presentation** only 5 out of 64 submissions selected
- P. W. Chang, K. W. Huang, **J. Fagin**, J. Chan, J. Lin, *LenSiam: Self-Supervised Learning on Strong Gravitational Lens Images*, NeurIPS 2023 AI for Science Workshop.

TALKS

- Invited seminar to CUNY Lehman College, Bronx NY, U.S *May 2025*
- Invited seminar to Andrés Bello National University, Santiago, Chile *March 2025*
- Catching supermassive black holes with Rubin-LSST, Turin, Italy *July 2024*
- AstroAI Workshop 2024, Harvard-Smithsonian, Cambridge MA, U.S. *June 2024*
- Invited talk to Peter Melchior's group, Princeton University, Princeton NJ, U.S. *April 2024*
- The Restless Nature of AGN: 10 years later, Napoli, Italy *June 2023*
- Strong Lensing in the Era of Big Data, Otranto, Italy *June 2023*
- ICLR 2023 Workshop on Physics for ML, Kigali, Rwanda (virtual) *May 2023*
 - **Spotlight presentation** only 5 out of 64 submissions selected
- GothamFest, Flatiron Institute, New York NY, U.S. *January 2023*
- Lensing Odyssey 2022, Kouremenos, Crete, Greece *September 2022*
- Lensing Odyssey 2021, Kouremenos, Crete, Greece *September 2021*
- American Physical Society, Washington D.C., U.S. (virtual) *April 2020*

AWARDS

- Fee waiver, flights, and stipend to “Catching supermassive black holes with Rubin-LSST” conference
- Conference fee waiver and housing to “AstroAI Workshop 2024”
- Travel grant of €350 to “Strong Lensing in the Era of Big Data” conference

TEACHING EXPERIENCE

Taught 9 lab sections at CUNY Lehman College with typical class sizes of 20-25 students.

- Mechanics lab: Fall 2021, Spring 2022, Fall 2023 (calculus based)
- Electricity and magnetism lab: Fall 2021
- Physics of sound lab: Spring 2022, Fall 2022 $\times 2$, Spring 2023 $\times 2$

MENTORSHIP

- Co-mentoring 3rd year CUNY Ph.D. student Sophia Miskiewicz in machine learning for quasar microlensing caustic crossing predictions and microlensing simulations
- Co-mentoring 2nd year Universidad Andrés Bello Ph.D. student Matias Espinoza in machine learning for strongly lensed quasar and galaxy spectra reconstruction and redshift estimation

SCIENTIFIC COLLABORATIONS

- **Co-leading** a quasar variability task force in the LSST AGN science collaboration *2024 - Present*
- Member of the LSST dark energy science collaboration *2024 - Present*
- Member of the LSST AGN science collaboration *2021 - Present*
- Member of the LSST strong lensing science collaboration *2021 - Present*

SERVICES

- Co-founder of the astronomy journal club at the American Museum of Natural History
- Reviewer for MNRAS